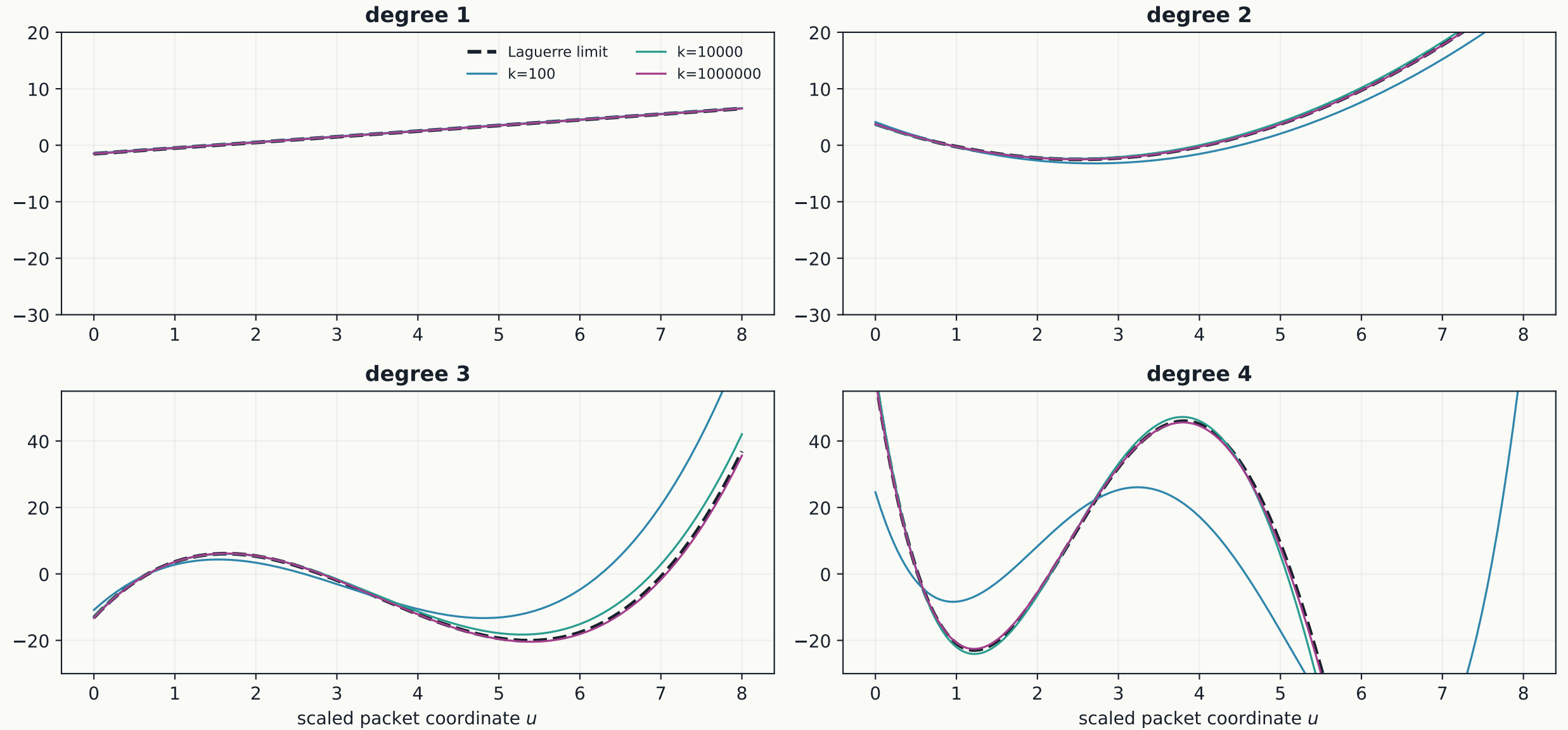Fixed-degree packet polynomials converge to monic generalized Laguerre polynomials